NIALL CASSIDY

INVESTIGATION

A2 WJEC COMPUTER SCIENCE

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Section 2: Investigation (10/100 marks

• Made full use of a range of appropriate methods to investigate the existing system.

• Completed a thorough investigation of the current system.

• Carried out extensive desk-based research into existing solutions to similar problems and identified the best features and facilities of these solutions to include in their solution to their chosen problem.

• Described the involvement of all stakeholders in the current system and described their requirements for the proposed project.

• Fully analysed data collected for input.

• Analysed in detail the processing carried out by the existing system.

• Given full consideration to all current system outputs.

• Fully considered and explained the limitations of the current system.

• Produced a working specification that clearly summarises the purpose of the project.

• Explained, with technical justification, the methods to be used in the solution.

• Detailed a range of objectives that include success criteria and clearly define the required outcomes.

2.0 Made full use of a range of appropriate methods to investigate the existing system.

2.1 Investigation of existing system using appropriate methods

An investigation into Mulligan Motors current system is beneficial as it will give me a deeper insight into the operations of their firm. I hope to be able to use this information to ensure the database system I create will match the exact needs of the staff and find any subtler user requirements that may have been missed during the discussion and feedback stage of the project, refining the specification in the process. In this investigation, I aim to find the strength and weaknesses of the system especially relative to my proposed system so I can reflect on the improvements I can make for the employees of Mulligan Motors

Identification of appropriate methods

Before continuing with the investigation, I decided it would be ideal if I identified the methods of investigation that I could use to maximise information retrieval whilst minimising the distraction of the investigation for Mulligan's Motors management which would bring them away from their business.

My initial approach for examining the present system will involve **observing** the staff and management during its operation. Ideally, this should be conducted discreetly, so that employees remain unaware they are being watched. The main benefit of this method is that it allows me to see how the business typically functions day to day, rather than how they might want me to perceive it. Nevertheless, this technique has some drawbacks, primarily that it only provides a glimpse into certain activities within the business, such as how they deal with customers using the current system, and doesn't cover, for instance, data management.

My second observation method will be an online customer survey. This will involve asking customers to fill out a survey, highlighting areas of the business they believe need improvement. This will be the only part of the investigation that involves customer input. Whilst this offers a unique viewpoint on the system, customers have no incentive to complete the survey and the response rate and, consequently, sample size, may be too low to paint a complete picture of the system from the customers point of view.

My third investigative method will involve conducting a **structured interview**. This will be a one-on-one session with an employee of Mulligan Motors experienced with the current system using a list of questions I will have beforehand. This approach provides an opportunity to gain a deeper understanding of how the system works and to gather the staff member's opinions on its strengths and weaknesses. However, this method is the most time-consuming of those listed and may be biased by the specific views of the interviewed staff member. Staff members of Mulligan's Motors may also exaggerate the responsibilities and actions they do to impress me which is another problematic issue with private interviews.

The fourth technique I will employ is a **staff questionnaire**. This will include a series of closed-ended questions, which will facilitate concise responses, making it simple and intuitive for the staff to complete, and for me to analyse, as the nature of these questions will streamline the organisational aspect of sorting and comparing responses on the same topic. However, this method also has limitations, chiefly that the brief nature of the questions makes it challenging to gather detailed insights about the current system from the responses given. Additionally, if the questions are too time consuming or complicated, I fear that the staff might purposely give false information to shorten their answers and get it the questionnaire done and over with. For this reason especially I will aim to keep the questions concise and close ended specifically

2.1.1: Observation

For my observation of the Mulligan Motors, I visited the store on set days across two weeks, ensuring that if the staff were having a bad week or any external events occurred, such as another dealership having a sale, during the first week I would still have an additional week to truly visualise Mulligan Motors under its normal operation. I asked the manager to not inform the employees and staff to help with this objective also. If the staff knew they were being formally watched, they might do their regular tasks differently to align their actions to be more 'by the book'. Over this period, I was to gain full access to the dealership floor inside and outside, as well as areas were the staff filled out forms and customers paid for the second hand cars that Mulligan Motors provides.
I noticed a first serious fault in the current traditional system of Mulligan Motors is that when customers asked to book a test drive, the staff had no records or capability to book a test drive for a customer. It was done on an informal, casual basis which is vulnerable to communication errors between employees. During the second day of my observation, a potential buyer was told by a member of staff that they could have a test drive at 2PM the next day no problem. However, when that same customer with a limited time schedule arrived at the dealership expecting a reserved timeslot for his test-drive, he found that another customer had taken the same car for a test drive with the permission of another employee. This resulted in a very disgruntled and justifiably confused customer asking why an employee of Mulligan Motors promised him a test drive only to give it to another customer. My electronic system would fix this issue by implementing the ability to reserve and book a test drive for a particular customer

which an employee would be able to quickly look at and ergo inform the customer times they can take a test drive.

Another problem I noticed during my observation at Mulligan Motors was that if a customer asked about the availability of particular car or vehicle; the employee had no fast method of which they could right then and there inform the customer. I found that almost 80% of employees dealt with this query by physically navigating the dealership and looking at the stock themselves. This method may work for the more experienced salespeople in the dealership who are well-versed in car manufacturers and their models but for the remaining 20% of newer salespeople, they used a much slower method. These newer employees still unfamiliar or unconfident in their abilities to identify specific car models and series, had to either ask another staff member or unlock the stock records cabinet on the far side of the store and manually scan the stock records. This is extremely inefficient and provides poor customer service by leaving the customer by themselves of an extended amount of time.

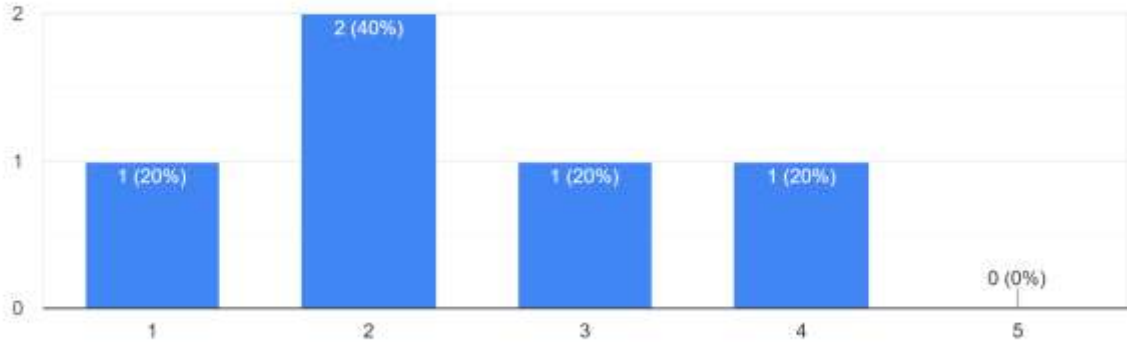
Lastly, I observed a significant issue with the way data was stored. Although the data was kept in a locked cabinet, all drawers could be unlocked with the same key, which was often left lying on or near the cabinet in predictable and unguarded places. This is a serious concern since the cabinet contains sensitive information about the business, its employees, and customers. Such lapses could allow any staff member to access the data, making it susceptible to theft or both accidental and malicious damage. Implementing a computerized system would address this issue by enabling different levels of access based on an employee's role as well as requiring a username and password to login. Encryption to safeguard the data from unauthorized access or destruction will also protect the firm’s data on my computerised system.

2.1.2 Method 2: Customer Survey

For my customer survey I decided the best method to ensure the highest possible response rate would to be that employees would ask customers at the end of their visit to Mulligan Motors to fill out a customer survey online. As a result of employees asking customers to fill out a quick five-minute survey, I received 5 responses after a day of waiting. The survey itself is composed of 8 questions with quick and simple quantitative questions asking the user to rate an experience from 1 – 10 at the beginning, then qualitive questions show up more at the end allowing the customer to give a more detailed review of their experience at Mulligan Motors. These responses are shown below.

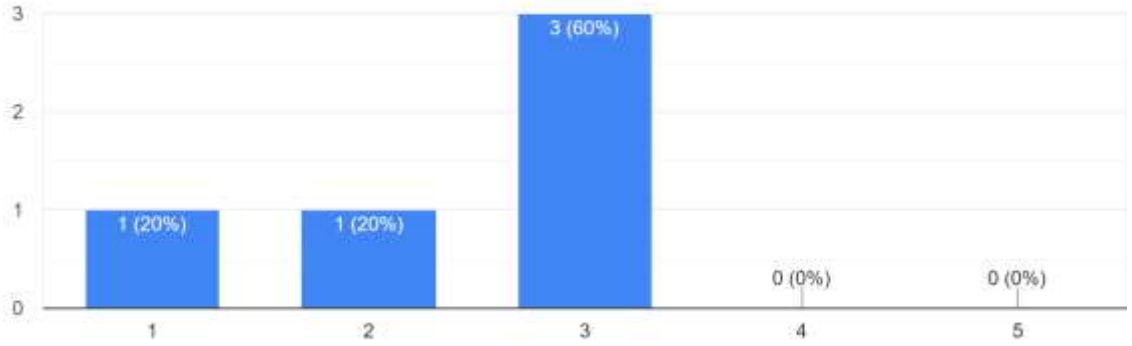
How would you rate your experience at Mulligan Motors today?

5 responses



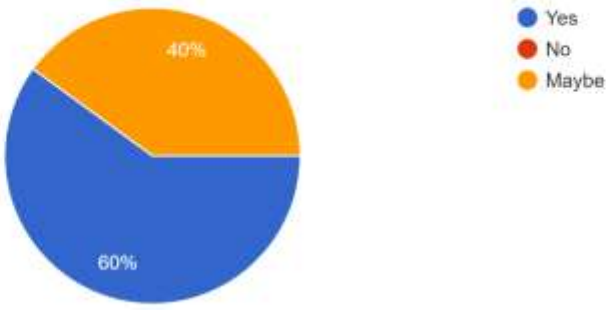
How would you rate your the speed of your service at Mulligan Motors today?

5 responses



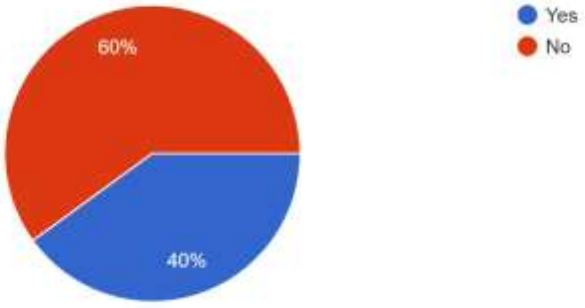
Do you think that the current system used in Mulligan Motors needs upgrading/modernized??

5 responses



Did you find the staff helpful and efficient during your visit?

5 responses



If you did not find the staff helpful or efficient, please explain why.

3 responses

I filled out a form for 20 minutes only for a new salesperson to spill coffee all over my paperwork

I wanted to book a test drive. The employee said this is not currently possible??

They seemed frustrated by their own system. I felt like a portion of that frustration was taken out on me

Was there anything you found particularly exceptional or enjoyable about your experience in Mulligan Motors?

5 responses

- The staff were very helpful in helping me pick a car.
- Leaving the building
- N/a
- Staff were eager and enthusiastic
- There was a sizeable selection of vehicles

Was there anything you found particularly disappointing or unenjoyable about your experience in Mulligan Motors?

5 responses

- The staff had lost the forms that they use to buy a car. Postponing my service by ~15 minutes.
- Entering the building
- Expected a more organised customer service
- N/a
- There was out of date information/ pricing on some select vehicles

Is there anything else you would like to add?

5 responses

The staff shouldn't need to go out and physically check the dealership to tell me what is. in stock.

The new intern was very slow at his job and seemed fu

A closer look at the details of a car available should be more easily available

Special discounts should be advertised more clearly

n/a

2.1.3 Method 3: Structured Interview.

For the sake of allowing the interviewee to be more focused, and making the interview overall more organised, i have decided to make the structure of the interview sectioned around three topics.

1. Data management.
2. Security.
3. System processes.

The following questions were asked by me and answered by the manager of Mulligan Motors

Data Management:

Q1: How is data currently stored?

Data is currently stored in locked wooden cabinets. If data is not confidential, such as a simple car details form, the staff will often leave it on the desk beside the office.

***Q2:* How would you edit or append a record?**

If a minor detail such as the DOB of a client or the make of car is incorrect, I will simply scribble it out and write the correct data beside it. If the majority of the data needs edited, I

would tend to just get a blank form out of the cupboard and start transferring the details across.

Q3: How is data sorted and organised in your traditional system?

Once a week we dedicate a staff member's shift to sorting, validating and correcting all the forms in our cabinet. This is a long process but allows our system to keep efficient and keep our customer service response times competitive with other auto dealerships in the area. However, no sorting is done at any other time; if a staff member wants to search the cabinet for a form 1-2 days before this process of sorting occurs, the cabinet can be extremely messy and difficult to navigate until it is sorted again.

Q4: How is new data entered onto the current system.

Data is written onto a paper form and inserted into the cabinet in the correct drawer. It is up to the employee to ensure their writing is legible and the information correct.

Q5: How is data deleted?

For minor changes, data is typically crossed out, whereas if an entire file needs to be removed, it's simply pulled from the cabinet and discarded.

Q6: Are there any measures in place to uphold data integrity?

Currently, we have a process in place to uphold data integrity. Every week, we assign a staff member to a shift specifically for sorting, validating, and correcting all the forms in our cabinet. In spite of this, this is an expensive and arduous process which cannot keep up with the rate of which of business is scaling.

Q7: Do you take any precautions to backup data in case of an event that destroys data?

We do not have a data backup plan in place at the moment. I believe the volume of data we have is too extensive to manually review and make copies of everything.

Q8: What do you think is the best way to upgrade Mulligan Motor's paper-based system?

Enhancing our system by transitioning to a digital format, instead of relying on paper, will upgrade our operations in several ways. First, it will reduce clutter in the staff area, making the workspace more organized. Additionally, it will eliminate redundant data when updating or adding files, ensuring efficiency. Finally, this shift will enable us to efficiently create backups, thereby keeping our data safe and ensuring customer satisfaction.

Security:

Q1: How is data kept secure?

Data is kept in a locked cabinet, accessible only to authorized staff. This helps in preventing unauthorized access and ensures that our customer and business data remain protected.

Q2: Who has access to the locked cabinet?

Access to the locked cabinet is restricted to certain staff members. These individuals are responsible for managing and maintaining the data stored within the cabinet.

Q3: Do you feel that these security measures of sufficient for Mulligan Motors?

Given the limitations of a paper-based system, while our current security measures offer a basic level of protection, they are not foolproof. Transitioning to a digital system would provide more robust security features, including encrypted data storage and controlled access.

Q4: What improvements do you think should be made?

To enhance our data security, we should consider transitioning to a digital system. This change would allow for better data management, secure backups, and reduced risk of data loss or unauthorized access.

Q5: What happens if the key to the cabinet is lost?

In the event that the key is lost, we would need to contact a locksmith to gain access to the cabinet and potentially replace the lock. This situation would cause significant disruption and delay in accessing our data, highlighting the need for a more secure and reliable system.

System processes:

Q1: Is it possible to book a test drive on the current system?

Currently, the booking of a test drive is not possible in our traditional system. We find it too time consuming and logistically challenging. Even if we wanted to, we do not currently have the forms on hand or space in our cabinet to even store the test drive bookings.

Q2: How would a staff member deal with a query asking for all the cars with a particular specification be shown?

A staff member would need to manually search through the cabinet for the forms that match the specified criteria. This process is time-consuming and prone to errors due to the manual nature of our current system.

Q3: How long does it usually take for a new staff member to become used to the current system?

It typically takes a new staff member a few weeks to become familiar with the current paper-based system. The learning curve involves understanding how data is organized, how to properly file and retrieve documents, and the procedures for maintaining data accuracy.

Q4: Do you believe that how the system currently handles processes negatively impacts the customer experience.

Yes, the manual nature of our current system can lead to delays and errors, which may negatively impact the customer experience. Losing customer contact information or compromised data integrity on certain car information can upset our clients significantly. Transitioning to a digital system would streamline processes, reduce wait times, and improve overall efficiency.

Q5: What system do you think would be the most efficient method of handling processes and queries in Mulligan Motors.

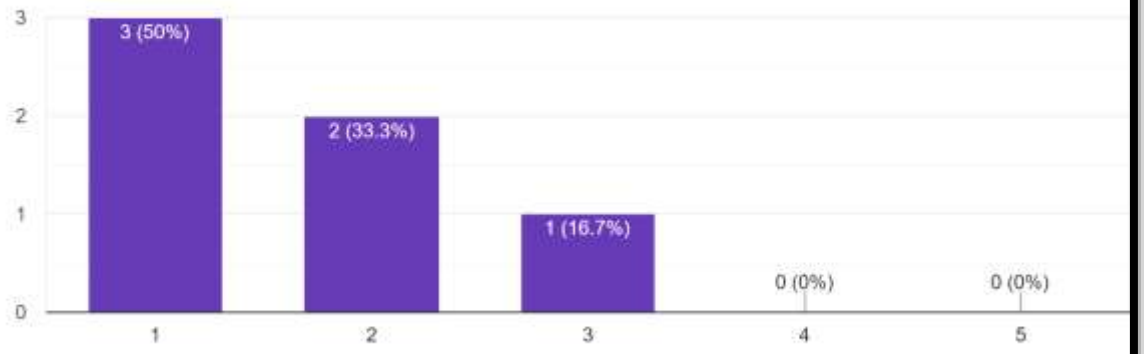
Implementing a digital data management system would be the most efficient method for handling processes and queries. It would allow for quick data retrieval, accurate record-keeping, and easier management of customer information, ultimately enhancing the customer experience and operational efficiency.

2.1.4 Method 4: Staff Questionnaire.

For my final method of investigation, my staff questionnaire will be give an insight into the current system from those who have the most hands-on experience with it. Due to the staff of Mulligan Motors not having much free-time, I aimed to have the survey to be as easy to complete as possible. As a result of this objective, I settled on a format of google forms once again with all questions having a structure of 'How much do agree with the statement…' This allows the employees of the firm to stop and start the form easily in between breaks, and complete it anywhere due to the online aspect of the questionnaire. The following graphs will show the question and portray the amount who selected that answer. '1' represents strongly disagree, '3' is neutral, and '5' is strongly agree.

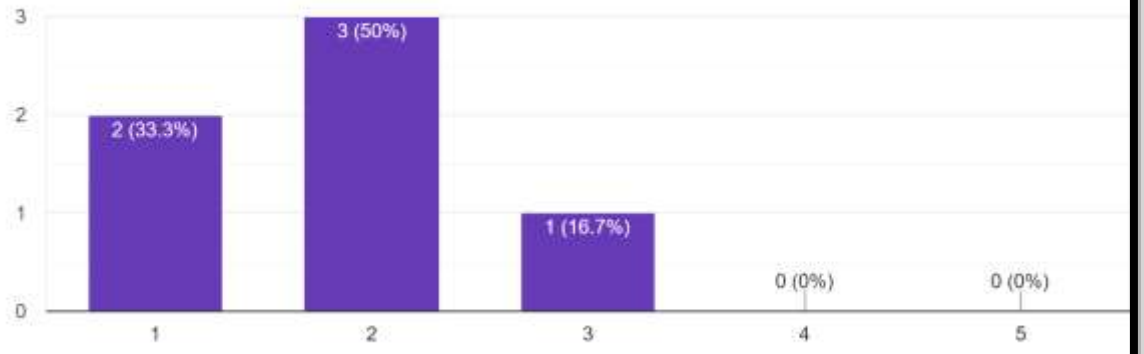
The current system is effective.

6 responses



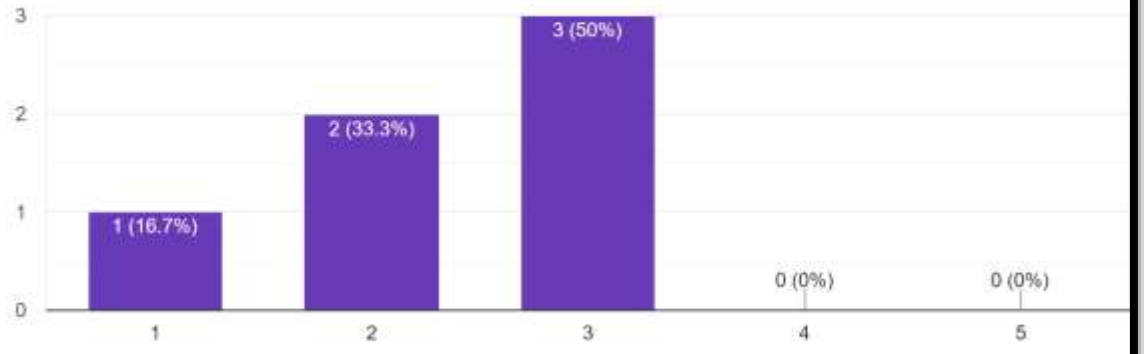
The current system is easy to learn.

6 responses



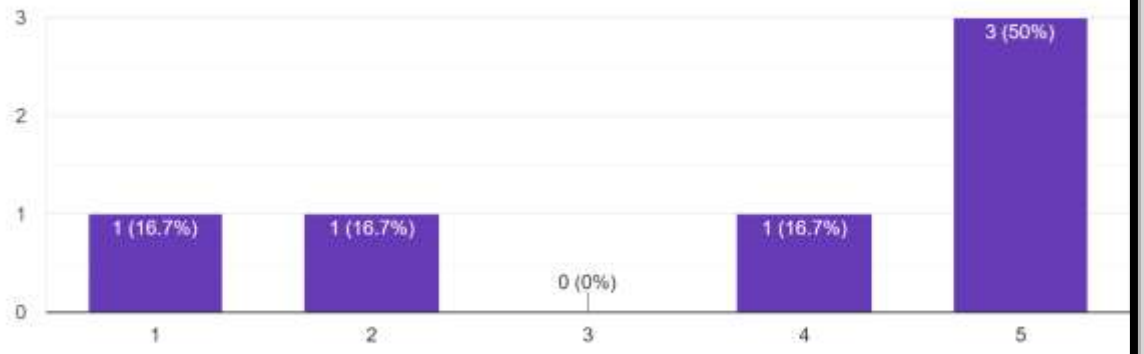
It is hard to make a mistake in the current system

6 responses



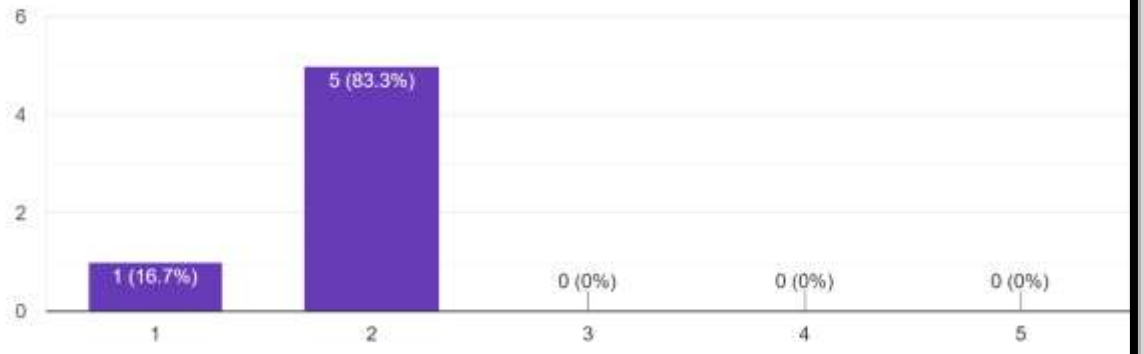
I wouldn't mind learning a modern digital system if it replaced the traditional system

6 responses



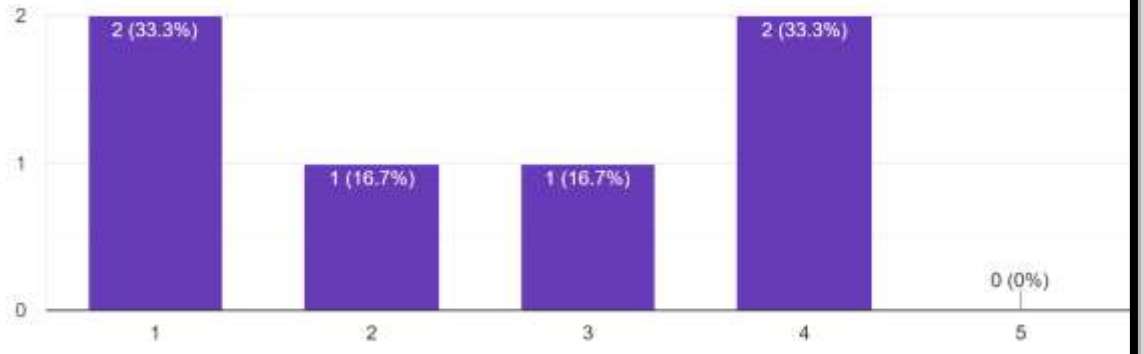
Records are rarely lost, damaged, or require replacing in the current system.

6 responses



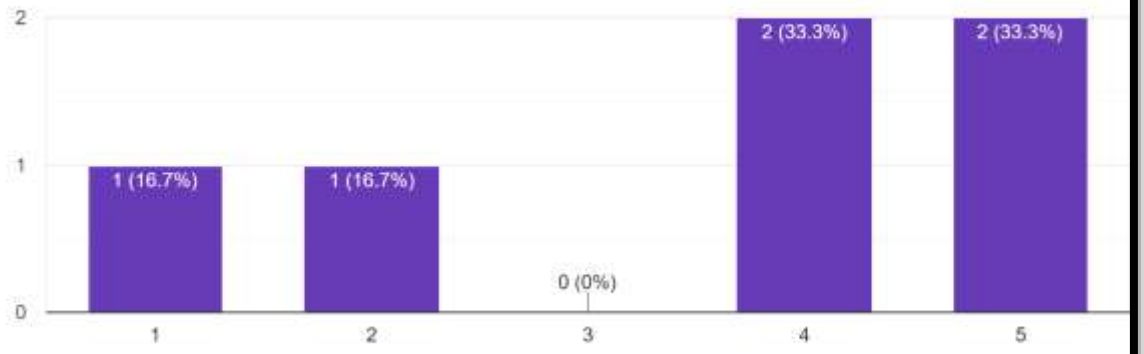
The current system is capable of providing outstanding customer service.

6 responses



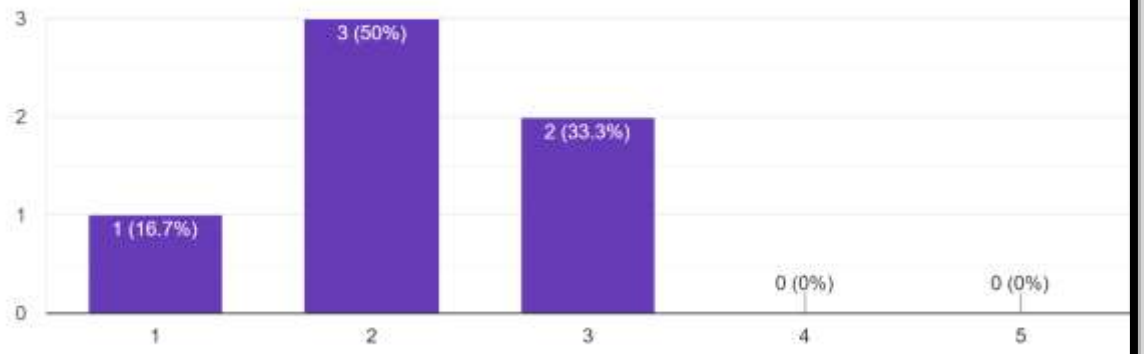
With the the current system, I feel capable of providing basic customer service

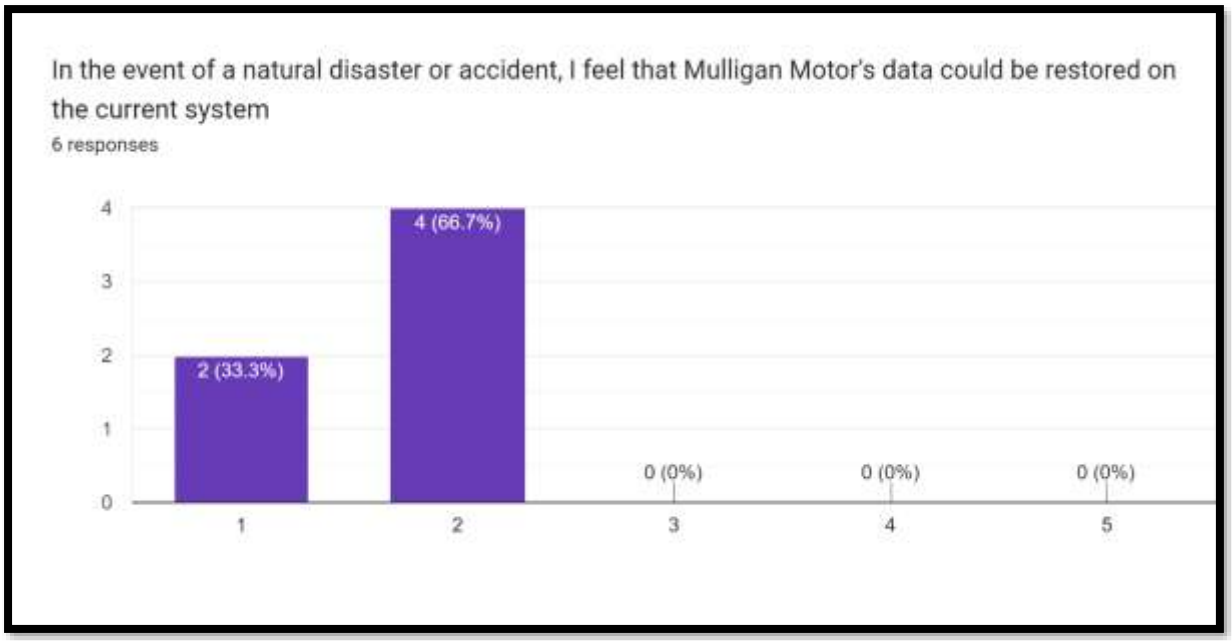
6 responses



I feel that data security measures in Mulligan Motors are sufficient with the current system.

6 responses





Evaluation of results

Q: The current system is effective.

The chart shows that the majority of staff (50%) rated the current system's effectiveness as a 1, indicating significant dissatisfaction. Another 33.3% rated it a 2, and 16.7% rated it a 3, with no ratings of 4 or 5. These low ratings suggest that the current paper-based system is inefficient and ineffective. Upgrading to the proposed electronic database would likely address these issues by streamlining operations, reducing errors, and improving overall effectiveness.

Q: The current system is easy to learn.

The general sentiment among staff is that the current system is not user-friendly, with the majority rating it below neutral. This indicates that the system is cumbersome and may benefit from a more intuitive and streamlined approach.

Q: It is hard to make a mistake in the current system

The chart shows mixed views on the current system's error-proneness, with 50% rating it as a 3 and 33.3% rating it as a 2, indicating some difficulty in avoiding mistakes. Only 16.7% rated it as a 1, showing that a portion of the staff finds the system prone to errors, highlighting the need for improvements.

Q: I wouldn't mind learning a modern digital system if it replaced the traditional system

The second chart reveals strong support for adopting a modern digital system, with 50% rating their willingness to learn as a 5. This indicates that despite some resistance, who are

likely senior staff who are not used to digital technology, most staff are open to transitioning to a more reliable electronic database.

Q: Records are rarely lost, damaged, or require replacing in the current system.

This question and it's answers reveal that a substantial majority of Mulligan Motors staff, 83.3%, disagree with the statement that records are rarely lost, damaged, or require replacing in the current system. Additionally, 16.7% of the staff strongly disagree, indicating severe dissatisfaction. This data evidenced my suspicions that Mulligan Motors' current record management system has significant issues with record loss, damage, and the need for frequent replacements. Notably, no respondents were neutral, agreed, or strongly agreed with the statement, underscoring a unanimous perception of inadequacy. Given this feedback, it is evident that Mulligan Motors should prioritize enhancing their record management system, this will be achieved in my proposed solution as with digitization and improved data management practices. This will mitigate these issues and improve operational efficiency and staff satisfaction

Q: The current system is capable of providing outstanding customer service.

The chart highlights that a majority of staff, 50%, feel that the current data security measures at Mulligan Motors are insufficient, as they selected "Disagree." Additionally, 16.7% strongly disagree with the sufficiency of the data security measures. While 33.3% of the respondents are neutral, no respondents agreed or strongly agreed with the statement, suggesting a general concern about the adequacy of data security. This feedback points to a clear need for improvements in the data security systems in place at Mulligan Motors to ensure the protection of sensitive information. By implementing these improvements, the proposed solution will significantly enhance data security at Mulligan Motors, addressing the concerns raised in the survey and ensuring sensitive information is better protected.

Q: With the the current system, I feel capable of providing basic customer service

The bar chart reveals mixed feedback on staff confidence in providing basic customer service with the current system at Mulligan Motors. While 33.3% of respondents agree and another 33.3% strongly agree that they feel capable, 16.7% strongly disagree and 16.7% disagree, indicating a significant portion of the staff finds the system inadequate. This mixed feedback suggests that while many staff members are comfortable with the current system, a notable minority experience difficulties. This highlights the need for system improvements to ensure all staff members feel equally capable of providing high-quality customer service. By implementing the proposed solution, which includes a more user-friendly interface and better access to customer and inventory information, Mulligan Motors will enhance the overall capability of staff to deliver excellent customer service.

Q: I feel that data security measures in Mulligan Motors are sufficient with the current system.

The bar chart reveals mixed feedback on staff confidence in providing basic customer service with the current system at Mulligan Motors. While 33.3% of respondents agree and another 33.3% strongly agree that they feel capable, 16.7% strongly disagree and 16.7% disagree, indicating a significant portion of the staff finds the system inadequate. This mixed feedback suggests that while many staff members are comfortable with the current system, a notable minority experience difficulties. This highlights the need for system improvements to ensure all staff members feel equally capable of providing high-quality customer service. By implementing the proposed solution, which includes a more user-friendly interface and better access to customer and inventory information, Mulligan Motors will enhance the overall capability of staff to deliver excellent customer service.

Q: In the event of a natural disaster or accident, I feel that Mulligan Motor's data could be restored on the current system

The chart reveals a significant lack of confidence among staff regarding the current system's ability to restore data in the event of a natural disaster or accident. A majority of respondents, 66.7%, disagree, while 33.3% strongly disagree with the statement. No respondents expressed neutrality or confidence in the system's data restoration capabilities.

This overwhelming lack of confidence indicates a critical vulnerability in Mulligan Motors' data management and disaster recovery processes. It underscores the urgent need for implementing a robust backup and data recovery system. By incorporating regular automated backups, offsite storage solutions, and a comprehensive disaster recovery plan, Mulligan Motors can ensure that data can be efficiently and effectively restored following any unforeseen events. These improvements will not only protect valuable data but also enhance overall operational resilience and staff confidence in the system.

2.7 Carried out extensive desk-based research into existing solutions to similar problems and identified the best features and facilities of these solutions to include in their solution to their chosen problem.

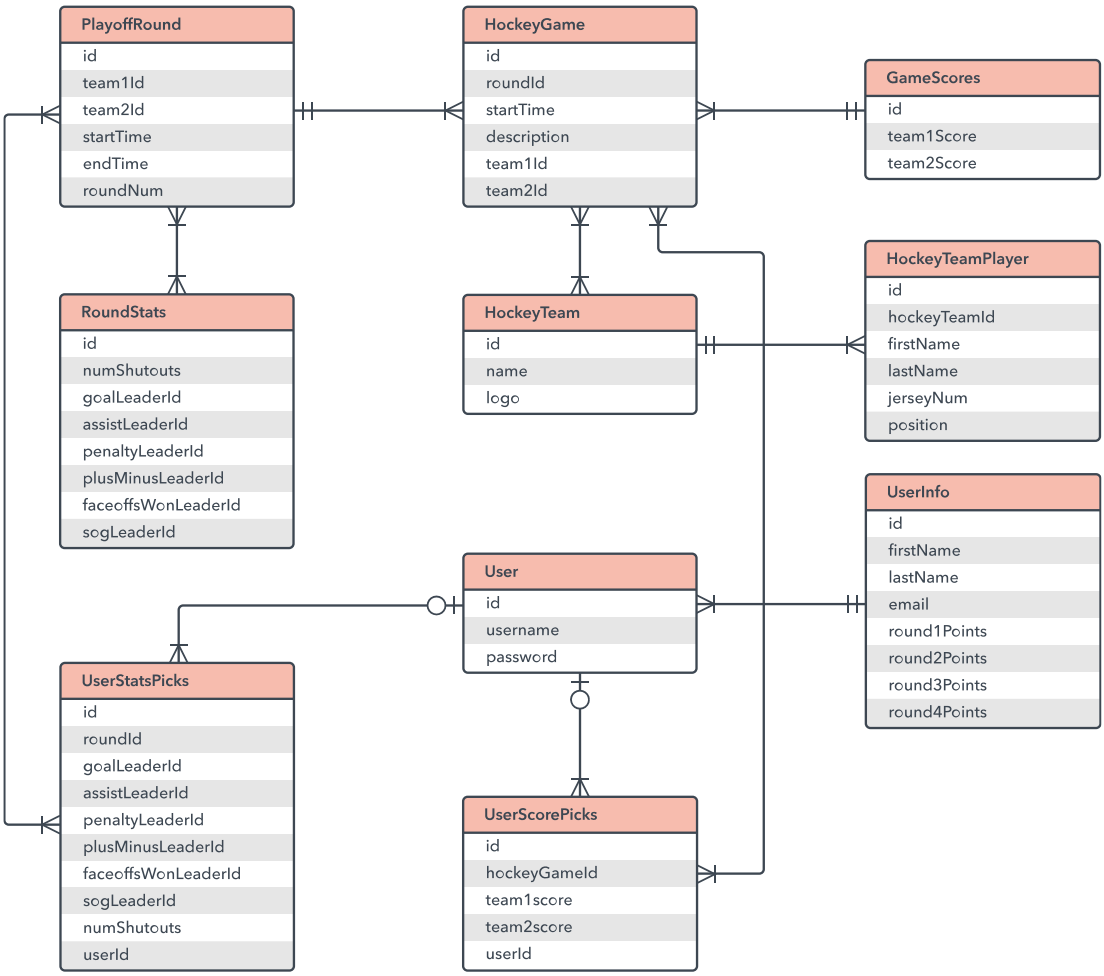
Existing solutions

To maximise consumer welfare from my system, I decided to evaluate and analyse other examples of user interfaces that I could find from the internet. By viewing numerous successful examples and understanding why they are effective, I hope to incorporate the effective designs into my system.



This simple, dual label login interface is a highly successful format for a login screen. This is evidenced by its almost universal use in all applications and businesses, big or small. This particular image is from Microsoft's Windows 10 login page. I believe this will be an effective

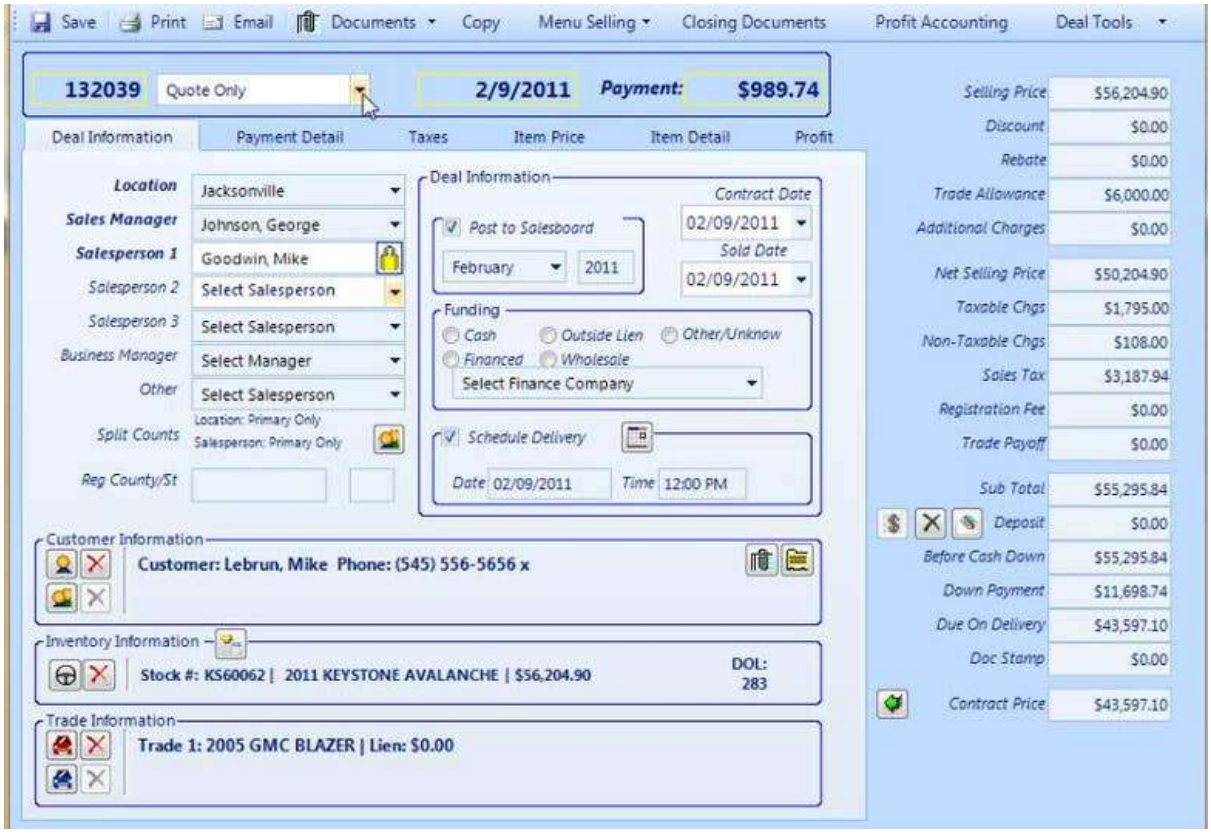
system for Mulligan Motor's system as it will be simple for me to design and create, putting little stress on the budget. In addition, this is straightforward and intuitive, reducing the training time for new employees or when the system is first released to Mulligan Motors. As part of the login page, I plan to implement a feature that will display characters inputted into the password label as asterisks (*). This will add a layer of security to the system as a fellow employee or customer cannot simply look over a staff members shoulder to gain access to the system using another's login information.



The diagram above is an example of an Entity Relationship Diagram (ERD), illustrating how different tables within a relational database are interconnected. Although this model is more extensive and complex than the database I will create, it offers valuable insights into the structure of a relational database system. Specifically, this example provided useful ideas on linking various tables within my database and understanding valid methods of establishing these connections, primarily through foreign keys.

Additionally, it highlights the types of relationships within a relational database, such as one-to-one and many-to-one relationships, and methods to avoid creating many-to-many relationships, which are not permitted. This is mainly achieved with an intermediate table, like 'HockeyGame' in the example. Although, this diagram does contain some many-to-many

relationships, which may limit its value as an example. Each table's fields also have a specified length and data type, a feature I plan to implement in my system.



The image above serves as an excellent example of a data entry screen, where data can be input and stored in a database. This screen uses some of the same inputs I plan to use in my own system, making it a useful model for designing a data entry interface. The input fields are clearly labelled to ensure accurate data entry. The buttons are also clearly labelled, allowing users to understand their functions easily.

Additionally, this example is well-designed, with related data, such as salesprice, grouped together. This organization makes the interface more user-friendly and easier to navigate. Aesthetically, the system provides a warm and welcoming interface which Mulligan Motors might appreciate to raise morale amongst employees using the system. The system also makes a point to not overwhelm the user with misleading images or information. The use of tabs at the top of the screen allows for additional functionality without cluttering the interface with too many buttons.

2.8 System inputs, processes, & outputs

The following is the data currently collected by Mulligan Motors, the processes they complete on the data, and the outputs they receive from the current system.

INPUTS:

- EmployeeID
- Employee Name
- Pay/hr(£)
- Vehicles sold
- Commission made(£)
- Contract type(0 Hour/Part-Time/Full-Time)
- Vehicle
- Vehicle type
- Vehicle Age
- Vehicle Manufacturer
- Vehicle Model
- Number of previous owners
- Vehicle Known issues
- Vehicle Price(£)

PROCESSES:

- Adding staff details
- Adding vehicle details
- Adding order details
- Editing/Appending data
- Deleting data
- Archiving old data
- Adding customer details

OUTPUTS:

- A list of vehicles stocked
- List of all staff members
- List of all placed orders
- List of all customers who have ordered an item

Evaluation of methods

1. **Observation**

I believe starting my investigative process with observation was an effective start to my process. Viewing Mulligan Motors in person allowed me to build an understanding of the whole process and give me a layer of knowledge on how this system acts in real life situations. It gave me a crucial foundation to being my other investigation methods, such as what questions to ask on the staff questionnaire or interview.

Viewing the practical problems of the firm also allowed me to have sample problems that I can test my own digital database solution against and compare how it would deal with the problem compared to the current paper-based system. This ensures that the user requirements I make are correct and are indeed met when I eventually hand over the system to the owners of Mulligan Motors.

However, whilst the observation of Mulligan Motors gives a fundamental overview of the company, I feel it lacked in the detail that it had given me. For example, certain employees had recognised that I was critically examining the system and started acting differently and taking longer to do certain tasks, such as filling out forms. This is likely due to them mistaking my system review for a employee review. Consequently, the staff started acting more, 'by the book' and focused on making less mistakes they could be criticised on.

2. Customer survey

The customer survey allowed me to collect qualitive and quantitative data about the satisfaction of those who were served by staff using the paper-based system. It gave me a different angle to identify the weaknesses and strengths of the current system. The results of the survey could definitely be improved upon with addition of a second survey when the new digital solution is implemented. Customers can compare their previous experiences with the ideally more efficient database system, contrasting the weaknesses and strengths of both.

3. Structured Interview

The interview process I felt gave me the most detailed view of the traditional system. Having all my questions answered by the manager himself on the spot allowed me to have developed responses with the opportunity to ask follow-up questions question if necessary. Insightful information such as data security, backups, and storage was shared during this interview will helped me understand the more technical details of the paper-based system. I felt that the interview likely could have been improved with thee addition of a group interview, or multiple interviews with different staff members. This would have allowed me to gain multiple answers and point of views which would have corroborated each other's answers.

4. Staff Questionnaire

The questionnaire was useful in that it allowed me to gain the understanding of the system from those who actually use it most. It gave me important opinions on

certain aspects of the document system. The lack of backups, security, and capability to provide outstanding customer service were highlighted in the survey with a majority of respondents signalling they were unhappy with the currents system's features of the current system. Furthermore, I found that the question,' I wouldn't mind learning a new system digital system if it replaced the current system', was especially important. It actually indicated that staff are willing to train in the proposed solution should it be implemented. Fortunately, almost all staff universally said they strongly agree with the statement, otherwise, the feasibility of the project would be put into question.

However, it should be noted that data collected via this means is vulnerable to being biased towards the staff members own interests and personal views. Since it is not an objective view of the system, other data should be taken into account using the other methods of investigation to form the design of the proposed solution for Mulligan Motors.

2.10 Limitations of current system

The limitations of the current paper-based system at Mulligan Motors are significant and multifaceted, affecting security, efficiency, data integrity, and overall business operations. I have decided it best to organise the most glaring problems before proceeding with the investigation.

Security risks are a major concern, as paper files can be easily misplaced or accessed by unauthorized personnel. Without proper access control, even low-level employees could alter or forge documents, and there is no way to trace unauthorized actions, leaving the company vulnerable to data tampering. The system also makes it difficult to comply with regulations like the Data Protection Act of 2018, risking fines or legal issues if personal data is mishandled.

The backup and disaster recovery process is labour-intensive, with staff needing to manually reprint and fill out forms, increasing the chances of error and delays. Additionally, paper records are susceptible to physical damage, such as fire or flooding, which could result in irreversible data loss. Data integrity is also compromised, as lost or misplaced records lead to redundancy, and mistakes or miscommunications cause inconsistent information. Paper forms are prone to issues like illegible handwriting or spills, further undermining the quality and accessibility of the data.

Finally, **Inefficienc**y is another key issue. Searching and sorting records manually is time-consuming and prone to human error. As the company grows, these inefficiencies will increase, leading to further delays and higher labour costs. The system's scalability is limited—manual processes will become increasingly difficult to manage, making the current

system unsustainable in the long run. Additionally, the paper-based system is resource-intensive, requiring significant storage space and employee time to manage.

2.11 Essential Stakeholders

Owner/Manager:

- The system should be user-friendly to minimize the need for extensive training for new employees.
- It should allow the restriction of access to sensitive data, such as customer personal information, pricing, and financial records, to authorized personnel only.
- A secure login system is necessary to protect the system from unauthorized access.
- The system should have facilities for creating and managing staff profiles, enabling easy onboarding and modification of employee details.
- The ability to generate financial reports, including sales performance and inventory data, is essential or making informed decisions.

Sales Staff:

- The system should have a functional, easy-to-use graphical user interface (GUI) to help sales staff quickly input and access customer information and vehicle details.
- Accurate recording of vehicle pricing is critical, and the system must allow sales staff to make necessary updates to the prices of vehicles when required.
- The ability to track customer interactions and sales leads, ensuring follow-up actions are efficiently managed.
- A search function for accessing customer and vehicle records quickly, ensuring no time is wasted during customer interactions.
- Access to inventory levels to check the availability of vehicles and parts, avoiding discrepancies.

Operational staff

- The system should enable efficient scheduling and tracking of vehicle maintenance and repairs.
- the system must provide real-time updates on parts and supplies.
- A user-friendly interface to streamline the recording of service and repair activities.
- A search function to find specific vehicles that need to be serviced.

2.12 Specification for proposed system

Taking into consideration all these concerns, I have devised a working specification for the project moving forward. However, this may need to change depending on the feasibility of the features mentioned. At present, the system should:

- Use a centralized database to store relevant information about each area of the dealership.
- Allow for the viewing, adding, deletion, and editing of stored data.
- Provide a secure login system to ensure the security of data.
- Use a system of access rights to restrict access to sensitive information.
- Use a GUI to present information in a visually pleasing and user-friendly manner.

2.13 Success Criteria

Staff

- Store employee records, including roles, performance metrics, and access rights.
- Search for staff members via StaffID or FirstName and LastName
- Ensure all data is appropriate and validated to a reasonable extent
- Generate a report that can be sorted alphabetically to show all staff members on payroll

Stock

- Maintain detailed records for vehicle inventory, including make, model, year, VIN, mileage, condition, price, and history.
- Create detailed vehicle profile pages specifications.
- Ensure all data is appropriate and validated to a reasonable extent
- Generate a report on all current items on stock, sorted alphebetically

Pay

- Ensure all data is appropriate and validated to a reasonable extent
- Be able to search payslips by date
- See total amount of payroll paid out in a certain month
- Generate a report to show all the payrolls given each month, sorted by date

Orders

- Log details of sales transactions, financing options, payment records, and contracts.
- Ensure all data is appropriate and validated to a reasonable extent
- Ability to search for customers via OrderID or stock sold

Customers

- Maintain customer records, including purchase history and contact details.
- Ability to search for customers via ID or First name and last name.
- Ensure all data is appropriate and validated to a reasonable extent
- Generate a report showing all customers currently registered on the system.

Overall

- Provide robust search and filtering options to locate specific records or vehicles.
- Restrict access to sensitive data based on user roles.
- Ensure the GUI is responsive and accessible.
- The database must be held in a SQL system.
- Ensure the system complies with relevant data protection regulations.

2.14 Methods to be used.

Now that a working specification has been drawn up for the system, I can decide on the methods to be used in developing the system and discuss why these methods are appropriate for Mulligan Motors.

The system will be programmed in Python, as this is the programming language in which I am most proficient, minimizing the amount of time needed to learn new syntax or concepts. Python source code is very easy to read, meaning that someone with little programming experience will be able to understand what the code does, increasing trust in the system. This simplicity also allows another developer who does not understand Python to read and comprehend the code.

The database engine used will be the SQLite3 module for Python. I have chosen this database engine as it allows for the creation and management of databases without requiring the installation of a database management system or database administrator, which minimizes the amount of setup needed to implement the system. It also allows for database manipulation within Python, enabling me to map database commands (e.g., add, delete, etc.) to buttons within the GUI. The database will store data in third normal form, reducing data redundancy and duplication, as well as optimizing performance.

The Human-Computer interface for this project will be a Graphical User Interface (GUI). These GUIs will be developed using the Tkinter module for Python, as this module allows for the construction of simple GUIs within Python, which does not have native support for building GUIs. I have chosen this method of building GUIs as it allows me to apply my skills within Python while minimizing the amount of new syntax and processes that I need to learn. Although the GUIs that Tkinter allows are quite simple, they provide enough functionality for this application. I will prioritise functionality over design and aesthetics as customers will not be interacting with the system and staff should not be impacted by a lack of design unlike the performance of this system.

In addition to the Python files and the database, I plan that a text file will be used to store the access key of the user currently logged into the system. This will be used to check whether they have access to the screen they are trying to view. Python's simple facilities for reading, writing, and appending to text files are sufficient due to the straightforward nature of their use.

The use of self-documenting identifiers will be a standard practice, making reviewing and debugging the code easier. Clear variable names and function descriptions will facilitate easier testing and maintenance.

A range of validation techniques will be employed during the development of this system and will be used on every piece of data to be entered into the system. This will include presence, lookup, range, and type checks to ensure that data entered is of the correct type for the attribute of the table it is being entered. This will make searching for and using data easier, as all data in the system will be of the correct type and format.

Several methods will be employed to maintain high code quality throughout the development process. Firstly, the use of global variables will be minimized, as their overuse can lead to performance issues. Any function that needs to be carried out multiple times across the program will be implemented as a function. This reduces the amount of code written, decreases file size, saves programming time, and makes the system easier to test since the function can be tested once instead of each implementation. I believe this is a particularly important point for Mulligan Motors as their budget for computing hardware is low, so low performance computers need to be able to handle to database system.

By following these methods, I hope to make the system for Mulligan Motors efficient, secure, and easy to maintain, ensuring that it meets the dealership's operational needs effectively.